بسم الله الرحمن الرحیم

Intelligent Analysis of Biomedical Images

First semester 1405–1406 (Fall 2026)

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Department of Computer Engineering

Homework 5

Due: 25 Dey 1405

- Submission deadline: **25 Dey 1405, 23:59**. Late submissions follow the policy announced in class.
- Submit your answers as a single PDF named IABI_Theory_HW05_[First-Name]_[Last-Name]_[Student-Id].pdf. Typed answers are preferred; handwritten answers must be scanned clearly and legibly.
- For each problem, the main papers are listed at the beginning. All concepts needed to solve the problem are stated in the question text; the papers are pointers for deeper reading. Using other sources is allowed but must be cited. Unsupported claims receive no full credit.
- Collaboration and the use of AI tools are allowed if disclosed clearly; the final answer must be your own work and you must be able to defend it. If you collaborated, name your collaborators at the start of your answer.
- The points of each part are printed at its beginning. The theory total is **100** points, plus up to **10** bonus points. Short, precise, well-argued answers are preferred over long generic ones.
- In calculation problems, write the steps and the units. A final number without justification receives no full credit.
- In paper-based parts, distinguish what the paper shows from what it claims. Pay attention to the data split, the baseline, the metric, and external validation.
- If anything is ambiguous, ask on the course forum (Quera) rather than messaging the TAs directly.
- **Homework designer:** Amirhossein Izadi

Theory (100 + 10 points)

Problem 1. From U-Net to nnU-Net: geometry, loss, and validation in dense segmentation [17]

Background. A hospital network wants to segment liver tumors in contrast-enhanced computed tomography (CT). The network has three centers: the first acquires thin slices, the second thick and strongly anisotropic slices, and the third uses a different contrast-injection protocol. The research team first proposes a plain U-Net, then Attention U-Net, and finally nnU-Net. The question is which model and which evaluation criterion fit this study, and what a single Dice number can hide. The question starts from model geometry, moves to the loss, and ends with why "correct configuration derived from the data" matters more than "a new architecture".

Main sources: U-Net, Attention U-Net, nnU-Net, nnU-Net Revisited. All concepts needed are stated in the text; the papers are pointers for deeper reading.

(a) [4 points] Receptive field and information flow. Consider a U-Net with four encoder blocks. Each block applies two 3×3 convolutions followed by a 2×2 max-pooling. The bottleneck contains two more 3×3 convolutions. Padding keeps the spatial size unchanged inside convolutions.

1. Compute the receptive field of one unit in the bottleneck. Hint: each 3×3 convolution adds $2 \times$ jump; each 2×2 pooling adds $1 \times$ jump and then doubles the jump.
2. Why can one decoder unit see both the large bottleneck context and the high-resolution boundary detail of a skip connection? Write one sentence about the effective receptive field.
3. Explain the difference between 2×2 max-pooling and a 3×3 convolution with stride 2, in terms of lost information and aliasing.
4. Name one cost and one benefit of a skip connection (other than segmentation accuracy).

(b) [4 points] Loss under severe class imbalance. Assume that only 0.5% of the pixels belong to tumor. For each pixel, $p_i = \sigma(z_i)$ and the label is $y_i \in \{0, 1\}$.

1. Derive the BCE gradient with respect to z_k. With an approximate calculation for a 512×512 image, show why the model can reduce the loss by predicting "all background".
2. Write the smoothed Dice loss with $\epsilon = 10^{-6}$ and compute its gradient with respect to z_k. Then explain why this gradient concentrates on predicted-positive pixels.
3. Propose a combined loss and state the role of each term. If the clinical goal is to reduce false negatives, which change in the loss or in the weighting do you recommend?
4. For $y = [1, 0]$ and $p = [0.1, 0.1]$ (which gives $A = \sum p_i y_i = 0.1$ and $B = \sum p_i + \sum y_i = 1.2$), compute the positive and negative components of the Dice gradient and their ratio. Repeat for $p = [0.9, 0.9]$. What do these two numbers say about the self-normalization of the gradient?

(c) [3 points] The attention gate and its interpretation. In Attention U-Net, the additive attention gate is defined as

$$q_{\text{att}} = \psi^T \sigma_1(W_x x_l + W_g g + b_g) + b_\psi, \qquad \alpha_l = \sigma_2(q_{\text{att}}), \qquad \hat{x}_l = \alpha_l \odot x_l,$$

where x_l are skip features and g is the gating signal from the decoder.

1. Explain the roles of σ_1 and σ_2, and why this module helps manage variability in organ shape and size.

2. For three spatial locations, let $q_{\text{att}} = [-1, 0, 2]$. If σ_2 is the sigmoid, compute the coefficients α_l. If $x_l = [1, 1, 1]$, what is the weighted output?

3. Can the map α_l be used as an "explanation" of the model? Give three reasons for doubt and state which additional test is required before claiming faithfulness (connect to Problem 3).

(d) [4 points] Configuration from data, not architecture: the nnU-Net lesson and the Revisited result. The nnU-Net rule is: the dataset fingerprint (voxel spacing, median shape, intensity distribution) decides, not the architect's taste — the target resampling spacing, the patch size derived from memory, and the cascade rule used when the whole context does not fit in one patch.

1. Resample the volume $512 \times 512 \times 128$ of the second center with spacing $0.7 \times 0.7 \times 5$ mm to $1 \times 1 \times 1$ mm: what are the new dimensions and volume? (Hint: $512 \cdot 0.7 = 358$ and $128 \cdot 5 = 640$.) Compute the ratio of the new volume to the old one. If the patch size stays fixed at 128^3, why does each patch cover less anatomical context, and which decision is required (change the target spacing or add a low-resolution cascade)?

2. The nnU-Net Revisited study reports that, when the nnU-Net baseline is run with the same tuning budget and the same preprocessing, the claimed gap of new architectures closes. Explain why this is the same lesson as item 1 (performance comes from configuration, not from a new block). Then write three audit questions that must be asked of any claim of being "better than nnU-Net" (baseline, tuning budget, test set).

3. Define Dice, HD95 and ASSD. Construct a geometric scenario in which Dice improves while HD95 worsens. Explain why this matters clinically for tumors adjacent to an organ at risk.

4. Why can a random slice-level split falsely inflate performance? Propose a correct data-split design for the three-center study and draw an evaluation table containing at least Dice, HD95, lesion-level sensitivity and performance broken down by tumor size. Why is a single mean insufficient?

(e) [2 points] Design decision. If your compute budget allows training only one model, which do you choose among a carefully tuned U-Net, Attention U-Net, and nnU-Net? In at most 5 lines, justify it with one number from (a)/(b) (e.g., the receptive field 140 or the positive-to-negative gradient ratio) and an argument about the inter-center gap and inference cost. An answer without a number receives no credit.

⋆ **[2 points].** How much memory does one 128^3 feature map with 30 channels in fp32 need? (Hint: $128^3 \approx 2.1\,\text{M}$.) If the encoder forward pass is about 670 MB, the full forward pass with the decoder about 2.7 GB, and the backward pass plus Adam state raises this to about 8–12 GB, why can the patch not be made larger than 128^3 on a 12 GB device? (Hint: volume scales with the cube of the patch length; $160^3/128^3 \approx 1.95$.)

Solution 1

(a)

1. Starting from $r = 1$ and jump $= 1$:

$$\begin{aligned}
\text{block 1: } r = 1 &\xrightarrow{3\times3} 3 \xrightarrow{3\times3} 5 \xrightarrow{\text{pool}} 6, && \text{jump} = 2,\\
\text{block 2: } r = 6 &\xrightarrow{3\times3} 10 \xrightarrow{3\times3} 14 \xrightarrow{\text{pool}} 16, && \text{jump} = 4,\\
\text{block 3: } r = 16 &\xrightarrow{3\times3} 24 \xrightarrow{3\times3} 32 \xrightarrow{\text{pool}} 36, && \text{jump} = 8,\\
\text{block 4: } r = 36 &\xrightarrow{3\times3} 52 \xrightarrow{3\times3} 68 \xrightarrow{\text{pool}} 76, && \text{jump} = 16,\\
\text{bottleneck: } r = 76 &\xrightarrow{3\times3} 108 \xrightarrow{3\times3} 140.
\end{aligned}$$

Hence the receptive field is 140×140 pixels. The arithmetic is: a 3×3 convolution adds $2\times$jump, and a 2×2 pooling adds $1\times$jump before doubling the jump. The final jump is 16, so the bottleneck sees 140 input pixels per side.

2. The bottleneck path supplies large context, while the skip path supplies high-resolution detail; the effective receptive field is a combination of the two, not a single number. The skip preserves boundaries and gradients, but it can also carry noise and artefacts into the decoder.

3. Max-pooling is a fixed, non-learnable operator and can discard fine information; a stride-2 convolution is learnable and can retain useful features, but without an adequate low-pass filter it produces aliasing. Both reduce resolution; the difference is which information is kept and whether the downsampling can adapt to the data.

4. Cost: extra memory and parameters in the decoder (and a wider feature stack at each resolution). Benefit: a shorter, more direct gradient path, hence easier optimization and faster convergence.

(b)

1. With $p_i = \sigma(z_i)$:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_i \left[y_i \log p_i + (1 - y_i)\log(1 - p_i)\right], \qquad \frac{\partial \mathcal{L}_{\text{BCE}}}{\partial z_i} = \frac{p_i - y_i}{N}.$$

For a 512×512 image, $N = 262{,}144$. With 0.5% foreground there are about $1{,}310$ positive and $260{,}834$ negative pixels. If the model predicts background everywhere ($p_i \approx 0$), the total contribution of positive pixels to the mean gradient is about $1{,}310/N \approx 0.005$, and the background pixels contribute no positive gradient. Predicting all background therefore lowers the loss; the optimum of a constant predictor is the base rate $c = 0.005$, with entropy $H(0.005) \approx 0.0315$ and exactly zero gradient. This is the flat local minimum that motivates a region-based loss.

2.

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon}, \qquad \epsilon = 10^{-6}.$$

With $A = \sum p_i y_i$ and $B = \sum p_i + \sum y_i$:

$$\frac{\partial \mathcal{L}_{\text{Dice}}}{\partial z_k} = -p_k(1-p_k)\,\frac{2y_k(B+\epsilon) - (2A+\epsilon)}{(B+\epsilon)^2}.$$

For a positive pixel with $p_k \approx 0$, $y_k B - A \approx B$, so the gradient is $\approx -2p_k/B$ and does not scale with the (huge) background count. For a negative pixel with $p_k \approx 0$, $y_k B - A = -A \approx 0$, so its contribution is almost zero. Dice therefore concentrates on predicted-positive pixels and on the boundary.

3. A common choice is

$$\mathcal{L} = \lambda \mathcal{L}_{\text{BCE}} + (1-\lambda)\mathcal{L}_{\text{Dice}}, \qquad \lambda \in [0.3, 0.5].$$

BCE provides stable pixel-level learning; Dice handles imbalance and overlap. To reduce false negatives, use Tversky with $\beta > \alpha$, up-weight positive pixels, or apply hard negative/positive mining.

4. Neglecting the 10^{-6} term: for $y = [1, 0]$, $p = [0.1, 0.1]$, $A = 0.1$, $B = 1.2$. The positive component is $-0.09(2 \cdot 1.2 - 0.2)/1.44 \approx -0.138$; the negative component is $-0.09(0 - 0.2)/1.44 \approx +0.0125$; the ratio of magnitudes is about 11. For $p = [0.9, 0.9]$, $A = 0.9$, $B = 2.8$: the positive component is $-0.09(5.6 - 1.8)/7.84 \approx -0.0436$, the negative one $+0.09(1.8)/7.84 \approx +0.0207$, ratio about 2.1. As the predicted volume grows, both components shrink (self-normalization by B); the gradient is normalized by the predicted size, not by the fixed N.

(c)

1. σ_1 is usually a ReLU and creates the non-linear interaction between skip features and the gate; σ_2 is a sigmoid and keeps the attention coefficient in $[0, 1]$. The gate uses the decoder signal (coarser semantic context) to emphasize regions relevant to the target organ and suppress background; this helps with organ deformation and size variation.

2. With $\sigma_2 =$ sigmoid:

$$\alpha = \left[\frac{1}{1+e^1}, \frac{1}{1+e^0}, \frac{1}{1+e^{-2}}\right] \approx [0.269, 0.5, 0.881].$$

For $x_l = [1, 1, 1]$, the weighted output is $[0.269, 0.5, 0.881]$.

3. The map α_l is an attention coefficient, not a causal explanation. Three reasons for doubt: (i) it can be high for irrelevant regions; (ii) it is optimized for segmentation, not for faithfulness to the decision; (iii) its insensitivity to input changes or to weight randomization is not guaranteed. A faithfulness claim needs the tests of Problem 3 (randomization, deletion/insertion, intervention).

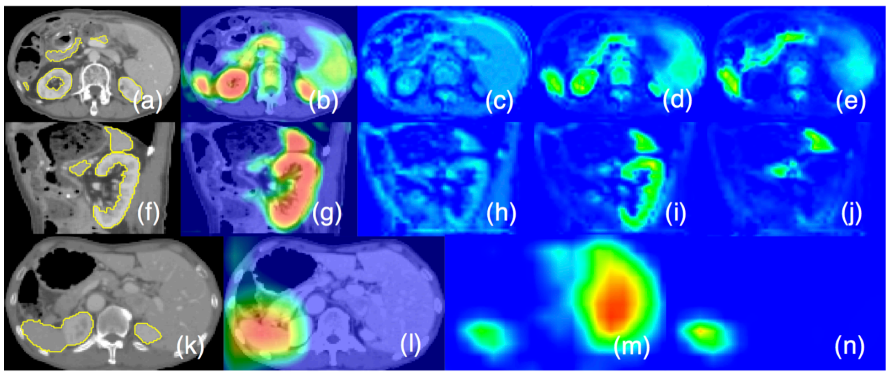


Figure: attention coefficients and skip-connection activations before and after gating in Attention U-Net. The gate suppresses irrelevant regions, but this suppression is not an explanation of the decision (Fig. 3 of Attention U-Net).

(d)

1. Resampling: $512 \cdot 0.7 = 358$, so the new dimensions are $358 \times 358 \times 640$. The old volume is $512 \cdot 512 \cdot 128 = 33{,}554{,}432 \approx 33.6\,\text{M}$; the new one is $358 \cdot 358 \cdot 640 = 82{,}024{,}960 \approx 82.0\,\text{M}$; the ratio is $82.02/33.55 \approx 2.44$. With a fixed $128^3 \approx 2.1\,\text{M}$ patch, each patch covers a smaller fraction of the organ after resampling (the effective context shrinks) while patch memory stays constant. One must either use a coarser target spacing (losing detail) or add a low-resolution/cascade path so that the whole context is visible. The post-resampling volume forces this decision; it is not a matter of taste.

2. Same lesson: if performance comes from configuration (spacing, patch, preprocessing, tuning budget), a comparison that does not give the baseline the same configuration does not measure architecture. Three audit questions: Was the baseline run with the same hyperparameter-search budget and the same preprocessing? Is the data split identical (patient-level, multi-center)? Is there an external test set, and is the difference statistically significant?

3. Dice is volumetric overlap: $2|A \cap B|/(|A| + |B|)$. HD95 is the 95th percentile of the symmetric surface distance, and ASSD is the symmetric mean surface distance. Example: let the ground truth be a sphere of radius 10. Prediction A is a sphere of radius 9 (Dice about 0.84, HD95 about 1). Prediction B is the same radius-10 sphere plus a thin distant spike of length 30 voxels. Dice of B is almost 1 because the spike volume is negligible, but HD95 of B is about 30. Dice therefore improves while HD95 worsens. Near an organ at risk, a distant spike can damage critical tissue; Dice does not see this.

4. A random slice split puts slices of the same patient in both training and test; the model can memorize the patient and obtain inflated performance. Correct design: split at the patient level and hold out one center as external test. The table should contain at least center, tumor size, number of cases, Dice, HD95, lesion-level sensitivity and confidence intervals. A simple mean can be misleading when the subgroup mix changes (see Problem 6).

(e) The answer is graded. Rubric: one point for an explicit model choice backed by one number from (a)/(b) (e.g., receptive field 140, the 11-fold gradient ratio, or the 0.033 background trap), and one point for the argument about the inter-center gap and inference cost. No number, no credit. A good answer usually chooses nnU-Net (a strong baseline with automatic preprocessing) unless inference budget forbids it. Sample full answer: "nnU-Net; the 140 receptive field covers the whole tumor and the 11-fold gradient ratio neutralizes the background trap; the single training budget is spent on identical preprocessing of the three centers and 5-fold training, not on a new architecture."

$\star$ **answer.** $128^3 = 2{,}097{,}152 \approx 2.1\,\text{M}$; $2.1\text{M} \cdot 30 \cdot 4 \approx 252\,\text{MB}$ for one map. With a 160^3 patch the volume ratio is $(160/128)^3 = 1.25^3 \approx 1.95$; the $2.7\,\text{GB}$ forward pass becomes $\approx 5.3\,\text{GB}$ and the total 8–$12\,\text{GB}$ becomes ≈ 16–$23\,\text{GB}$, beyond $12\,\text{GB}$. Memory is therefore the binding constraint on patch size. Rubric: one point for $252\,\text{MB}$, one point for the cubic-scaling argument.

Problem 2. Detection: from R-CNN to DETR and YOLO — assignment and evaluation [17]

Background. A lung-cancer screening program wants to detect small pulmonary nodules in computed tomography (CT). The clinical constraint is at most one false positive per patient on average, while sensitivity for nodules smaller than 6 mm must remain high. Your team must choose among a two-stage detector (Faster R-CNN), a one-stage detector (YOLO), and a set-based detector (DETR). The question is about this choice and about what each metric hides.

Main sources: Faster R-CNN, YOLOv1, DETR, Deformable DETR, DINO, YOLOv10, PK-YOLO. All concepts needed are stated in the text; the papers are pointers for deeper reading.

(a) [4 points] Faster R-CNN and anchors.

1. Consider a 64×64 feature map with 3 scales and 3 aspect ratios per location. Compute the total number of anchors.

2. Explain the IoU-based assignment rule: when is an anchor positive, negative, or ignored? Why can a 5 mm nodule have no positive anchor at all? (Hint: for a 12×12 lesion against the smallest 128^2 anchor, compute the maximum possible IoU and compare it with the 0.3 threshold.)

3. The region proposal network (RPN) has two outputs: an object/background score per anchor and a box-coordinate refinement. Write the RPN loss and explain the roles of N_{cls}, N_{reg} and λ. Why is smooth L1 used for box regression?

4. How does RoIAlign differ from RoIPool, and why does the difference matter for small lesions?

(b) [3 points] YOLOv1 and small-object crowding.

1. For an image with a 19×19 grid, two boxes per cell and 20 classes, compute the output tensor dimensions. Explain the general formula $S \times S \times (B \cdot 5 + C)$.

2. Write the original YOLOv1 loss (coordinate, objectness and class terms). What is the role of the "no-object" term?

3. What happens if two small lymph nodes fall into the same grid cell? Which YOLOv1 mechanism causes one of them to be lost?

4. YOLOv10 is trained with two heads: a one-to-many head (rich supervision for the backbone) and a one-to-one head (consistent with the ranking criterion); at inference only the one-to-one head remains, so non-maximum suppression (NMS) is unnecessary. Explain why this lowers latency but does not fully solve the "two small objects in one cell" problem.

(c) [4 points] DETR and bipartite matching.

1. Explain the "set prediction" idea in DETR. Why does DETR not need non-maximum suppression?

2. Consider the cost matrix below for two predictions and two ground-truth boxes:

$$C = \begin{pmatrix} 0.2 & 0.9 \\ 0.8 & 0.3 \end{pmatrix}.$$

Find the optimal Hungarian matching and compute the total cost. Then solve the matrix

$$C = \begin{pmatrix} 0.40 & 0.41 \\ 0.41 & 0.40 \end{pmatrix}$$

as well: how large is the winning margin? If entry C_{00} is perturbed by $+0.03$, what does the assignment become? What does this say about assignment stability for two adjacent lesions of equal size?

3. Write the DETR loss and explain why its convergence is slower than that of Faster R-CNN.

4. Which part of this problem do Deformable DETR, DINO and Co-DETR each target? For a $512 \times 512 \times 128$ volume, analyse the cost of global attention in one sentence.

(d) [4 points] Metrics and the clinical decision. Suppose that over three patients we have five ground-truth boxes and six predictions ordered by score as follows (at IoU $= 0.5$). Predictions D and A are in patient 1, B and C in patient 2, and E and F in patient 3; each patient has at least one ground-truth box: respectively 2, 2 and 1 boxes.

Prediction	D	A	B	C	E	F
Score	0.95	0.90	0.80	0.70	0.60	0.50
Status	TP	TP	FP	TP	FP	TP

1. Build the precision–recall curve and compute AP with all-point interpolation.

2. Explain the difference between AP, "sensitivity at one false positive per patient", and FROC. Which is more appropriate for a screening program, and why?

3. Show how AP can hide a size bias. Propose an evaluation table broken down by small-/medium/large lesions.

4. If three radiologists mark different boxes, what does "ground truth" mean? Propose two ways to report inter-reader variability.

(e) [2 points] Deployment decision. With a budget of at most one false positive per patient, which do you deploy among Faster R-CNN, YOLO and DETR? In at most 6 lines, justify it with one number from (a)/(c) (e.g., the IoU bound for a small lesion or the assignment margin) and an explicit position on non-maximum suppression. An answer without a number and without a position on NMS receives no credit.

⋆ **[2 points].** With the criterion $m = s^{\gamma} \cdot p^{\alpha} \cdot \mathrm{IoU}^{\beta}$ and the conditions $\gamma_{o2o} = r\gamma_{o2m}$, $\alpha_{o2o} = r\alpha_{o2m}$, $\beta_{o2o} = r\beta_{o2m}$, show that the one-to-one and one-to-many heads produce the same ranking. For $r = 1$, $\gamma = 1$, $\alpha = 0.5$ and $\beta = 6.0$, write the exact equality. Why does dropping the one-to-many head at inference not reduce model capacity?

Solution 2

(a)

1. $64 \times 64 \times (3 \cdot 3) = 64 \cdot 64 \cdot 9 = 36{,}864$ anchors. With stride 16 on a 1000×600 image the grid is about 60×40, i.e. $21{,}600$ anchors before removing those crossing the border, of which about $6{,}000$ remain for training.

2. An anchor is positive if its IoU with a ground-truth box exceeds the high threshold (e.g. 0.7); negative if it is below the low threshold (e.g. 0.3); between the two it is ignored. A $5\,\text{mm}$ nodule can be smaller than the smallest anchor and obtain no sufficient IoU, so it is never counted as positive. Numerical hint: for a 12×12 lesion against a 128^2 anchor, even at the best alignment IoU $\leq 144/16{,}384 \approx 0.0088 \ll 0.3$; no anchor reaches 0.7, and only the rule "highest IoU with each ground-truth box" guarantees a positive.

3.

$$\mathcal{L}_{\text{RPN}} = \frac{1}{N_{cls}} \sum_i L_{cls}(p_i, p_i^*) + \lambda \frac{1}{N_{reg}} \sum_i p_i^* L_{reg}(t_i, t_i^*).$$

N_{cls} and N_{reg} normalize the two terms so that they have comparable scale; λ balances them (with $N_{cls} = 256$, $N_{reg} \approx 2400$, $\lambda = 10$ compensates the ratio $2400/256 \approx 9.4$). Smooth L1 is linear for large errors and quadratic for small ones, hence more robust to outliers than L2.

4. RoIPool rounds coordinates and introduces misalignment; RoIAlign samples with bilinear interpolation. For small lesions a one-voxel shift can be a large fraction of the mask.

(b)

1. $S = 19$, $B = 2$, $C = 20$: $19 \cdot 19 \cdot (2 \cdot 5 + 20) = 361 \cdot 30 = 10{,}830$.

2.

$$\begin{aligned}\mathcal{L}_{\text{YOLO}} = \lambda_{coord} \sum_{i,j} \mathbf{1}_{ij}^{obj} \left[(x - \hat{x})^2 + (y - \hat{y})^2 + (\sqrt{w} - \sqrt{\hat{w}})^2 + (\sqrt{h} - \sqrt{\hat{h}})^2 \right] \\ + \sum_{i,j} \mathbf{1}_{ij}^{obj} (1 - \hat{c})^2 + \lambda_{noobj} \sum_{i,j} \mathbf{1}_{ij}^{noobj} \hat{c}^2 + \sum_i \mathbf{1}_i^{obj} \sum_c (p_i(c) - \hat{p}_i(c))^2.\end{aligned}$$

The no-object term penalizes high confidence in cells that contain no object.

3. Only the box with the highest IoU with an object becomes "responsible"; a second object in the same cell is ignored and becomes a false negative. This is serious in crowded lymph nodes or small cells.

4. The two heads share the ranking criterion and therefore provide consistent supervision: the one-to-many head enriches the backbone, and the one-to-one head learns unique selection from the same features. YOLOv10 trains both with consistent dual assignments and removes NMS at inference. This lowers latency but does not fully solve "several small objects in one cell"; higher resolution, FPN or center-based assignment is still needed.

(c)

1. DETR maps a fixed set of N queries in parallel to class and box, and bipartite matching assigns each ground-truth box to a unique query. Since each object has exactly one responsible prediction, duplicate predictions are suppressed and NMS is unnecessary.

2. Option $1 \to 1, 2 \to 2$: $0.2 + 0.3 = 0.5$. Option $1 \to 2, 2 \to 1$: $0.9 + 0.8 = 1.7$. The optimal matching is diagonal. For the near-tie matrix, $\min\{0.80, 0.82\} = 0.80$ (diagonal) with a margin of only 0.02; with a $+0.03$ perturbation of C_{00}, $\min\{0.83, 0.82\} = 0.82$ and the assignment flips to the anti-diagonal. Two adjacent lesions of equal size therefore have a fragile assignment: a small perturbation changes responsibility and sends gradients to the wrong query.

3.

$$\mathcal{L}_{\text{Hungarian}} = \sum_{i=1}^{N} \Big[-\mathbf{1}_{c_i \neq \varnothing} \log \hat{p}_{\sigma(i)}(c_i) - \lambda_{\emptyset} \mathbf{1}_{c_i = \varnothing} \log \hat{p}_{\sigma(i)}(\emptyset) + \mathbf{1}_{c_i \neq \varnothing} L_{box}(b_i, \hat{b}_{\sigma(i)}) \Big],$$

where $c_i = \varnothing$ denotes a query with no object and $\lambda_{\emptyset}$ is the weight of the no-object class. The classification term for no-object queries is necessary. Convergence is slow because supervision is very sparse: only a few queries per image receive a strong gradient, and the matching is unstable early in training.

4. Deformable DETR improves cost and convergence with multi-scale deformable attention and local sampling; DINO stabilizes training with denoising and improved anchors; Co-DETR adds denser supervision through auxiliary one-to-many heads. For a $512 \times 512 \times 128$ volume, global attention over all voxels costs $O(N^2)$ with a very large N; local/deformable attention or slice-wise processing is required.

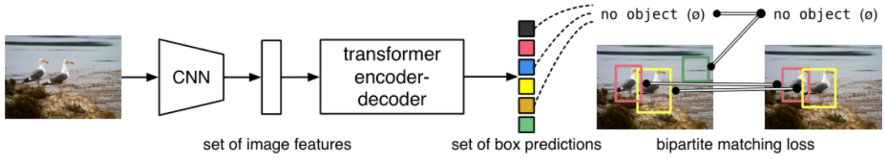


Figure: DETR predicts the final set of detections in parallel, and bipartite matching assigns each ground-truth box to a unique query; duplicate predictions are therefore suppressed and non-maximum suppression is unnecessary (Fig. 1 of DETR).

(d)

1. Order: D (TP), A (TP), B (FP), C (TP), E (FP), F (TP). With the patient assignment, at each threshold the number of false positives per patient is the total FP count divided by 3. The (R, P) points are $(0.2, 1)$, $(0.4, 1)$, $(0.4, 2/3)$, $(0.6, 0.75)$, $(0.6, 0.6)$, $(0.8, 2/3)$. With all-point interpolation:

$$AP = 0.2 \cdot 1 + 0.2 \cdot 1 + 0.2 \cdot 0.75 + 0.2 \cdot 0.667 \approx 0.683.$$

2. AP is the area under the PR curve across IoU thresholds; "sensitivity at one FP per patient" is directly tied to clinical burden; FROC plots sensitivity against FP per patient. For screening, FROC or sensitivity at a fixed FP rate is more appropriate because the cost of a false positive is explicit.

3. AP averages over all sizes; if a model is good on large lesions and poor on small ones, AP can remain high. Table: lesion size (small/medium/large) $\times$ count $\times$ AP $\times$ sensitivity at FP $= 1$.

4. "Ground truth" is a convention, not an absolute fact. Two options: (i) multiple readers and report inter-reader agreement (e.g. pairwise Dice); (ii) soft labels or probabilistic masks and evaluation against each reader.

(e) The answer is graded. Rubric: one point for a model choice backed by one number from (a)/(c) (the IoU bound for a small lesion, the 0.02 assignment margin, or $36{,}864$ anchors) and one point for an explicit position on NMS. No number and no position, no credit. Sample full answer: "DETR without NMS; the 0.0088 bound shows that standard anchors cannot see a small nodule, and the 0.02 margin makes hand assignment fragile; one-to-one matching removes the hand-tuned NMS threshold that either deletes a nearby nodule or keeps a false positive under a one-FP budget."

$\star$ **answer.** With $m = s^{\gamma} p^{\alpha} \mathrm{IoU}^{\beta}$ and the scaling conditions,

$$m_{o2o} = s^{\gamma_{o2o}} p^{\alpha_{o2o}} \mathrm{IoU}^{\beta_{o2o}} = s^{r\gamma_{o2m}} p^{r\alpha_{o2m}} \mathrm{IoU}^{r\beta_{o2m}} = \left(s^{\gamma_{o2m}} p^{\alpha_{o2m}} \mathrm{IoU}^{\beta_{o2m}} \right)^{r} = m_{o2m}^{r}.$$

For $r = 1$, $\gamma = 1$, $\alpha = 0.5$, $\beta = 6.0$, both heads equal $s\, p^{0.5}\, \mathrm{IoU}^{6.0}$, so the ranking is identical and the supervision is consistent. Because the knowledge sits in the shared weights, dropping the one-to-many head at inference does not reduce capacity. Rubric: one point for the equality, one point for the capacity argument.

Problem 3. Interpretability: is a saliency map the reason of the model? [17]

Background. A pneumonia classifier on chest radiographs reaches AUROC 0.92 on an in-distribution test set, but its AUROC drops to 0.58 at another hospital. Its Grad-CAM map highlights the lower lung zones. Is the model looking at the real pathology, at a shortcut, or at both? This question asks you to design an evaluation study that separates the two.
Main sources: Sanity Checks for Saliency Maps, Grad-CAM, Influence Functions, Assessing the (Un)Trustworthiness of Saliency Maps, The Clever Hans Mirage. All concepts needed are stated in the text; the papers are pointers for deeper reading.

(a) [4 points] From gradients to Grad-CAM and the sanity test.

1. Define three methods: vanilla gradient, Guided Backprop and Grad-CAM. Write the Guided Backprop recursion and explain why this method can resemble an edge detector.

2. Define the Grad-CAM weights as $\alpha_k^c = \frac{1}{Z}\sum_{i,j}\frac{\partial y^c}{\partial A_{ij}^k}$ and explain why a final ReLU is applied to the combined map.

3. Explain the cascading-randomization test: starting from the output layer and moving towards the input, randomize the weights of one layer at a time. Propose a quantitative criterion for comparing the original and randomized maps. (Hint: why can similarity computed with absolute values hide failure?)

4. If an explanation method remains almost unchanged after full weight randomization, what do you conclude? What does this imply for its claim of faithfulness?

(b) [4 points] Influence functions and data debugging.

1. Suppose a training sample z is added with weight ϵ. Using the first-order optimality condition, derive the upweighting influence:

$$I_{up}(z, z_{test}) = -\nabla_\theta L(z_{test}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_\theta L(z, \hat{\theta}), \qquad H_{\hat{\theta}} = \frac{1}{n}\sum_i \nabla_\theta^2 L(z_i, \hat{\theta}).$$

2. Why is computing $H^{-1}v$ difficult for large networks? Explain two approximations (e.g. conjugate gradient and LiSSA).

3. We have a mislabeled scan in the training set. How do you find it with influence functions? After finding it, which empirical test is needed to confirm it?

4. Name three serious limitations of influence functions (think of non-convexity, SGD noise, distribution shift and computational cost). Is this method a "causal proof" or a "screening tool"?

(c) [3 points] Shortcuts, causal graphs and the misleading map.

1. Draw a causal graph with Y (pathology), X (image), S (center/scanner) and M (marker or text on the image). Show how M can be correlated with Y without being its cause.

2. Explain why empirical risk minimization (ERM) can learn M instead of Y. Propose a "shortcut-removal" test that exposes this dependence.

3. Why can Grad-CAM be convincing and misleading at the same time? Give a concrete clinical scenario.

4. When the model is evaluated at a new center with low AUROC, what is the difference between "performance drop due to distribution shift" and "drop due to reliance on a shortcut"? How do you separate them?

(d) [4 points] Faithfulness versus plausibility.

1. Define the deletion/insertion curves. For a medical image, what do you delete and which quantity do you measure?

2. ROAD is a corrected deletion evaluation that progressively replaces important pixels with noisy, in-painted values to reduce artefacts and shape leakage. How does it differ from deleting the top-k pixels? Why must random deletion and importance-based deletion both be reported?

3. Define the pointing game for a segmentation model. Name two ways to cheat or bias this metric.

4. Design an evaluation protocol for the question: "Does the attention map indicate a region that is necessary and effective for the model's decision?" Your protocol must contain at least one randomization test, one intervention test, and one quantitative metric.

5. In three sentences, explain why a "clear heat map" alone is not evidence for clinical trust.

(e) [2 points] Decision rule. For the claim "this map is reliable enough for clinical deployment", name three required outputs: the result of the randomization test, the result of the intervention test, and one quantitative metric. In at most 6 lines, state the accept/reject threshold for each. An answer without a quantitative metric and a threshold receives no credit.

$\star$ **[1 point].** For linear regression with $L(z, \theta) = \frac{1}{2}(y - x^T\theta)^2$, write the gradient $\nabla_\theta L(z, \hat{\theta})$ and the Hessian $H = X^T X/n$, and make $I_{up}(z, z_{test})$ explicit. What does its sign say about whether sample z helps or hurts the prediction of z_{test}?

Solution 3

(a)

1. Vanilla gradient: $S(x) = \nabla_x f_c(x)$. Guided Backprop: during backpropagation the signal passes only through ReLUs whose forward activation and backward gradient are both positive:

$$R^l = \mathbf{1}_{\{R^{l+1}>0\}} \odot \mathbf{1}_{\{a^l>0\}} \odot (W^{l+1})^T R^{l+1}.$$

This removes negative signals, and the result resembles an edge detector.

2. Grad-CAM:

$$\alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k}, \qquad L^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The final ReLU keeps only features with a positive influence on the class.

3. Test: store the trained model; obtain the original map; randomize the weights layer by layer from output to input, recomputing the map after each step. Criterion: SSIM or rank correlation (Spearman) between the original map and the map after randomization. A faithful method must change rapidly as the randomization depth increases. Absolute-value caveat: the sign of the gradient (positive versus negative evidence) is part of the claim; two opposite maps become identical after taking absolute values, and their similarity stays high. The criterion must therefore be computed without absolute values, otherwise failure is hidden.

4. If the map remains almost constant after full randomization, the method depends mainly on the input/edges rather than on model parameters; its faithfulness to the model's decision is therefore in doubt.

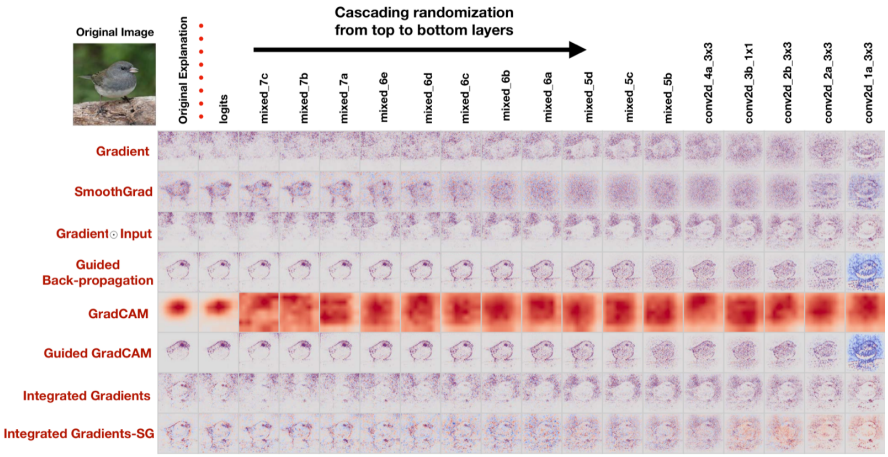


Figure: cascading randomization. From left to right more weights are randomized, and a faithful method's map should collapse. A stable map means the explanation is unrelated to the model (Fig. 2 of Sanity Checks).

(b)

1. Adding $\epsilon L(z, \theta)$ and using first-order optimality:

$$\hat{\theta}_{\epsilon,z} = \hat{\theta} - \epsilon H_{\hat{\theta}}^{-1} \nabla_\theta L(z, \hat{\theta}) + o(\epsilon),$$

and by the chain rule:

$$I_{up}(z, z_{test}) = -\nabla_\theta L(z_{test}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_\theta L(z, \hat{\theta}).$$

2. H is a matrix that is impractical to store and invert for large networks. Conjugate gradient solves $Hv = \nabla L(z)$ with repeated matrix–vector products; LiSSA approximates $H^{-1}v$ with a Neumann expansion and sampling of Hessian terms. Both need damping and an iteration-count setting.

3. First compute the influence of each suspicious sample on the wrong prediction; samples with a large positive influence are candidates for a wrong label. To confirm, correct the label or remove the sample and retrain; if the test prediction improves, the hypothesis is confirmed.

4. Limitations: (i) non-convexity and dependence on the optimum; (ii) SGD noise and Hessian instability; (iii) distribution shift and the influence not generalizing outside the set; (iv) computational cost. Influence functions are a screening tool, not a causal proof.

(c)

1. Causal-graph edges:

$$\begin{aligned} &S \to X, \qquad S \to M, \qquad M \to X, \qquad Y \to X, \\ &S \to Y, \qquad X \to \hat{Y}, \qquad Y \to \hat{Y}. \end{aligned}$$

M can be a scanner marker, side text or a chest tube. If M is correlated with Y in the training data, the model can use it as a shortcut.

2. ERM sees only empirical correlation; if M is an easier predictor than Y, minimizing the loss prefers it. Shortcut-removal test: erase M from the image (inpainting) or randomize it, and measure the change in prediction. If the prediction changes drastically, the model depends on M.

3. Example: a pneumonia model relies on a chest tube or the "PORTABLE" marker in the image; Grad-CAM highlights that region and the physician believes it, while the model never looked at the pathology.

4. The drop at a new center can come from a real distribution shift (e.g. image quality) or from reliance on a shortcut. To separate them: (i) the shortcut-removal test; (ii) evaluation on a set where the shortcut has been removed; (iii) patient-level correlation between prediction and M.

(d)

1. In the deletion curve, the k most important pixels are removed and the drop in class probability is measured; in the insertion curve, the same pixels are added to a blurred image and the increase is measured. For a medical image, "deletion" should use local averaging or inpainting so that the deletion itself does not create new artefacts.

2. ROAD measures importance by progressive deletion and progressive replacement of important pixels with noisy, in-painted values, and re-evaluates the same model; retraining is not required by the test definition; it also reduces the effect of pixel correlation. Simple top-k deletion can fragment a region and create artefacts. Reporting both (importance-based and random deletion) shows whether the map is better than chance.

3. Pointing game: checks whether the maximum of the attention map falls inside the lesion. Two biases: (i) larger/smoother models may always highlight the image center; (ii) if the lesion is large, any point inside it wins. Lesion size and its distribution must be controlled.

4. Protocol: (i) cascading weight randomization and measurement of map change; (ii) deletion/insertion with inpainting, compared with random deletion; (iii) input intervention: remove the highlighted region and measure the prediction drop; (iv) repeat across centers and readers. Quantitative metric: the difference in prediction drop between the highlighted region and a random region, with a confidence interval.

5. A heat map shows only where the model is sensitive, not why it decides; it can also be produced by a shortcut and is unstable under weight randomization. Clinical trust requires interventional evidence and reproducibility.

(e) The answer is graded. Rubric: half a point for each of the three outputs (randomization, intervention, quantitative metric) and half a point for an explicit accept/reject threshold. An answer without a quantitative metric gets at most half credit. A good answer: (i) the no-absolute-value similarity after full randomization must fall below the random-pair threshold; (ii) removing the highlighted region by inpainting must cause a significant drop, while random removal must not; (iii) report localization AUPRC against the mean-map baseline. Sample full answer: "reject; the no-absolute similarity remains high after randomization (edge-driven, not weight-driven), and removing the highlighted region by inpainting does not cause a significant drop; the map is not a justification for deployment, and a dedicated localization model is needed."

$\star$ **answer.** With $L = \frac{1}{2}(y - x^T\theta)^2$: $\nabla_\theta L(z, \hat{\theta}) = -(y - x^T\hat{\theta})x = -r_z x_z$ and $H = X^T X/n$. Hence $I_{up}(z, z_{test}) = -r_{test} r_z \, x_{test}^T H^{-1} x_z$. If the residuals have the same sign and $x_{test}^T H^{-1} x_z > 0$, then $I_{up} < 0$, i.e. upweighting z reduces the test loss (it helps); otherwise it hurts. Rubric: half a point for the formula, half for the sign interpretation.

Problem 4. Diffusion models: from noise to data, and from data to decisions [17]

Background. A research team wants to do two things: (i) generate synthetic brain MRI volumes to augment the training set of a tumor segmentation model, and (ii) reconstruct undersampled MRI using a generative prior. They first propose DDPM, then latent diffusion, and finally diffusion posterior sampling. This question asks you to examine the chain noise $\rightarrow$ data $\rightarrow$ decision both mathematically and from the point of view of clinical evaluation.
Main sources: DDPM, Score SDE, DDIM, Latent Diffusion, DPS, 3D MedDiffusion, CoMedBench. All concepts needed are stated in the text; the papers are pointers for deeper reading.

(a) [4 points] Forward and reverse processes. The forward process is defined as

$$q(x_t \mid x_{t-1}) = \mathcal{N}\Big(\sqrt{1-\beta_t}\, x_{t-1},\, \beta_t I\Big), \qquad \alpha_t = 1-\beta_t, \qquad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s.$$

1. Prove by induction that

$$q(x_t \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0,\, (1-\bar{\alpha}_t) I\big)\,.$$

2. Using Bayes' rule, derive the reverse distribution $q(x_{t-1} \mid x_t, x_0)$ and write its mean $\tilde{\mu}_t$ and variance $\tilde{\beta}_t$.

3. For $\beta_1 = 0.1$ and $\beta_2 = 0.2$, compute $\bar{\alpha}_2$ and write x_2 in terms of x_0 and one standard normal noise.

4. Compute the signal-to-noise ratio (SNR), assuming $\mathrm{Var}(x_0) = 1$, for these two steps and interpret it.

(b) [4 points] ELBO, score matching and Tweedie's formula.

1. By decomposing the likelihood over the chain $x_{0:T}$, write the evidence lower bound (ELBO) of the diffusion model and show that the KL term between $q(x_{t-1} \mid x_t, x_0)$ and $p_\theta(x_{t-1} \mid x_t)$ becomes a mean-squared error between $\tilde{\mu}_t$ and μ_θ.

2. With the parametrization $\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\Big(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\Big)$, show that the simplified objective becomes

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0,\epsilon,t}\left\|\epsilon - \epsilon_\theta\big(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\big)\right\|_2^2.$$

(Numerical note: the numbers reported in the DDPM paper are for CIFAR-10, not for this MRI problem: FID changes from 13.51 to 3.17 and the likelihood from 3.70 to 3.75 bits/dimension. Explain why transferring them directly to medicine is invalid and what domain-specific evaluation is required.)

3. Write the relation between ϵ_θ and the score $\nabla_{x_t} \log p_t(x_t)$, using the conditional expectation $\mathbb{E}[\epsilon \mid x_t]$. Then derive Tweedie's formula for $\mathbb{E}[x_0 \mid x_t]$.

4. Why is predicting ϵ better than predicting x_0 or $\tilde{\mu}_t$ directly, in terms of scale and gradient stability? (Think of large and small t.)

(c) [4 points] Latent diffusion and three-dimensional volumes.

1. Consider a $256 \times 256 \times 128$ volume and an autoencoder with downsampling factor $f = 8$ in every dimension. Compute the latent spatial dimensions and the volumetric compression factor. If the number of latent channels is 4, how many latent values are there in total?

2. Consider a small lesion with a diameter below 4 voxels. Explain why latent compression can erase it, and what remedies preserve the detail.

3. Explain classifier-free guidance with

$$\hat{\epsilon}(x_t, t, c) = \epsilon_\theta(x_t, t, \varnothing) + w\left(\epsilon_\theta(x_t, t, c) - \epsilon_\theta(x_t, t, \varnothing)\right).$$

   Describe the behaviour of samples for $w = 0$, $w = 1$ and $w \gg 1$.

4. Why is 3D medical generation harder than 2D generation? Explain three challenges (memory, anisotropy, anatomical consistency).

(d) [3 points] Evaluation of synthetic data and the inverse problem.

1. Define the three axes of synthetic medical-data evaluation: fidelity, downstream utility and privacy. Propose one quantitative metric per axis.

2. Design an experiment to answer: "Does adding synthetic data improve the Dice of a segmentation model on real, out-of-center data?" Specify the compute budget, the data split, and the statistical criteria.

3. For the inverse problem $y = Ax + \eta$, explain DPS and interpret the update

$$x_{t-1} = \mu_\theta(x_t, t) - \gamma_t \nabla_{x_t} \|y - A\hat{x}_0(x_t)\|_2^2 + \sigma_t z.$$

4. Explain the difference between measurement consistency and perceptual realism. Propose an evaluation that measures both, and show why a beautiful image can be clinically misleading.

(e) [2 points] The score field in a Gaussian example. For fixed $x_0 = 0$ and $\bar{\alpha} = 0.72$, compute the distribution p, the score $\nabla \log p$, and ϵ^*, and verify $\epsilon^* = -\sqrt{1 - \bar{\alpha}}\, \nabla \log p$. An answer without all three numbers receives no credit.

⋆ **[2 points].** Consider the sampler $x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\hat{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\,\epsilon_\theta + \sigma_t z$ with σ_t proportional to η. On the CIFAR-10 table with $S = 10$ steps, $\eta = 0$ gives 13.36 versus 41.07 for $\eta = 1$ (FID, lower is better). Which do you choose, and why? Explain the role of noise injection at small step counts with a variance argument.

Solution 4

(a)

1. Writing $x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1-\alpha_t}\epsilon_t$ and substituting recursively:

$$x_t = \sqrt{\alpha_t \alpha_{t-1} \cdots \alpha_1} x_0 + \sqrt{1 - \alpha_t \alpha_{t-1} \cdots \alpha_1}\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

With $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, the result is $q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1-\bar{\alpha}_t)I)$. For the induction, the recursive variance $V_t = \alpha_t V_{t-1} + (1-\alpha_t)$ with $V_1 = 1 - \alpha_1$ gives $V_t = 1 - \bar{\alpha}_t$.

2. By Bayes' rule,

$$q(x_{t-1}|x_t, x_0) \propto q(x_t|x_{t-1}) q(x_{t-1}|x_0),$$

and after completing the square:

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} x_t, \qquad \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t.$$

3. $\alpha_1 = 0.9$, $\alpha_2 = 0.8$, $\bar{\alpha}_2 = 0.72$. Therefore

$$x_2 = \sqrt{0.72}\, x_0 + \sqrt{0.28}\, \epsilon.$$

4. $\text{SNR}_t = \bar{\alpha}_t/(1-\bar{\alpha}_t)$. For $t = 1$: $0.9/0.1 = 9$; for $t = 2$: $0.72/0.28 \approx 2.57$. As t grows, the signal weakens and the noise dominates.

(b)

1. ELBO:

$$\mathcal{L} = \mathbb{E}_q \left[\log \frac{q(x_T|x_0)}{p(x_T)} + \sum_{t=2}^{T} \log \frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)} - \log p_\theta(x_0|x_1) \right].$$

If the two Gaussian variances are fixed and equal, the KL term is proportional to $\frac{1}{2\sigma_t^2}\|\tilde{\mu}_t - \mu_\theta\|^2$; under this assumption, absorbing the time weight into the objective gives a mean-squared error.

2. Substituting μ_θ and $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$, the term inside the norm simplifies to $\frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}}(\epsilon - \epsilon_\theta)$; absorbing the constant factor into the objective yields $\mathcal{L}_{\text{simple}}$. Numerical note (these numbers are from the CIFAR-10 experiment in DDPM, not MRI): dropping the theoretical weighting shifts capacity from low-noise steps (invisible detail) to intermediate steps (visible structure); FID improves from 13.51 to 3.17 at the cost of likelihood worsening from 3.70 to 3.75 bits/dimension. This is a perceptual-quality versus likelihood trade-off; FID alone is not a sufficient criterion for medical data and must be accompanied by domain metrics and downstream utility.

3. From $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$:

$$\nabla_{x_t} \log p_t(x_t) = -\frac{\mathbb{E}[\epsilon \mid x_t]}{\sqrt{1-\bar{\alpha}_t}}, \qquad \epsilon_\theta(x_t, t) \approx \mathbb{E}[\epsilon \mid x_t] = -\sqrt{1-\bar{\alpha}_t}\, \nabla_{x_t} \log p_t(x_t).$$

Tweedie's formula (with ϵ_θ estimating $\mathbb{E}[\epsilon \mid x_t]$):

$$\mathbb{E}[x_0|x_t] = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}.$$

4. For large t, x_t is almost pure noise and direct prediction of x_0 has very high variance; ϵ always has unit variance and the target has the same scale at every step. Predicting $\tilde{\mu}_t$ also depends on the noise schedule and its scale changes with t.

(c)

1. Latent dimensions: $256/8 = 32$, $256/8 = 32$, $128/8 = 16$, hence $32 \times 32 \times 16$. The volumetric compression factor is $8^3 = 512$. With 4 channels: $32 \cdot 32 \cdot 16 \cdot 4 = 65{,}536$ latent values versus $256 \cdot 256 \cdot 128 = 8{,}388{,}608$ voxels; ratio 128.

2. A lesion smaller than 4 voxels can dissolve into one latent voxel or merge with healthy tissue. Remedies: reduce f, use skip connections in the autoencoder, patch-based diffusion, or evaluate at the lesion level instead of global Dice.

3. $w = 0$: unconditional sampling; $w = 1$: standard conditional sampling; $w \gg 1$: stronger conditioning but low diversity and saturation/overcooking. In practice w is tuned between 1 and 7.

4. Memory: 3D attention or a full volume is expensive; anisotropy: slice spacing differs along z and convolutions/attention must model it; anatomical consistency: generating each slice independently can create inter-slice inconsistency.

(d)

1. **Fidelity:** FID/MMD, SSIM/PSNR, and radiologist assessment. **Downstream utility:** train a model on real + synthetic versus real only, with the same budget, and test on held-out real data. **Privacy:** membership inference, nearest-neighbour distance, duplication detection, and a DP guarantee when possible.

2. Design: patient-level and center-level split; three arms: real, real + synthetic, synthetic only; same architecture and training budget; evaluation on an external center; report Dice, HD95 and bootstrap confidence intervals. To prevent leakage, synthetic data must not be generated from test patients.

3. DPS uses a diffusion prior and a measurement model. At each step it first estimates $\hat{x}_0$ from x_t, then applies the gradient of the measurement inconsistency $\|y - A\hat{x}_0\|^2$ with coefficient γ_t, and adds noise. This drives the sample towards images that are consistent both with the generative prior and with the measurement.

4. "Measurement consistency" means $A\hat{x} \approx y$ with a small residual; "perceptual realism" means the image looks natural to a human. An image can be beautiful yet contain a structure that is absent from the measured data (hallucination). Evaluation must measure both separately: data residual, PSNR/SSIM, downstream Dice, and radiologist judgment.

(e) For fixed $x_0 = 0$ and $\bar{\alpha} = 0.72$: $p = \mathcal{N}(0, 0.28)$, because $x_t = \sqrt{0.72} \cdot 0 + \sqrt{0.28}\,\epsilon$. The score is $\nabla \log p = -x/0.28$ and $\epsilon^* = x/\sqrt{0.28}$. Verification:

$$-\sqrt{1-\bar{\alpha}}\,\nabla \log p = -\sqrt{0.28}\left(-\frac{x}{0.28}\right) = \frac{x}{\sqrt{0.28}} = \epsilon^*.$$

Rubric: half a point for each of the three numbers and half a point for the verification.

⋆ **answer.** These numbers come from the CIFAR-10 table of the DDIM paper; in that table $\eta = 0$ gives 13.36 and is the better choice. $\eta = 1$ injects fresh noise at every step, which accumulates when $S = 10$ is small (41.07); the deterministic path follows the same score field without injecting variance. With a small step budget, determinism is preferable; for MRI this choice must be confirmed by domain-specific evaluation and downstream utility. Rubric: one point for the choice with a number, one point for the variance argument.

Problem 5. Federated learning and differential privacy: collaboration without sharing data? [16]

Background. Five hospitals want to jointly train a pediatric tumor segmentation model. Two hospitals have rare subgroups, one has a different scanner, and none allows sharing raw data. The team first proposes FedAvg, then SCAFFOLD, and finally DP-SGD. The question is what each step buys and what it costs.

Main sources: FedAvg, SCAFFOLD, Federated nnU-Net, FL-MedSegBench, DP-SGD, RDP, MIA from First Principles, ADP-FL-MedSeg. All concepts needed are stated in the text; the papers are pointers for deeper reading.

(a) [4 points] FedAvg and client drift.

1. Write the FedAvg algorithm: the server sends w_t, each client performs E local steps, and the server takes a weighted average. Write the global update precisely.

2. Consider two clients with $F_1(w) = \frac{1}{2}a_1(w - b_1)^2$ and $F_2(w) = \frac{1}{2}a_2(w - b_2)^2$. With $a_1 = 1, b_1 = 0, a_2 = 4, b_2 = 10$, equal weights, $\eta = 0.1$ and $E = 2$, compute the weight after one FedAvg round. Compare it with one global gradient step and explain the client drift.

3. How does SCAFFOLD correct this drift with the control variates c_k and c? Write the update and explain why the correction vanishes when all clients have the same gradient at a point. (Hint: with the numbers of item 2, $c_1 \approx 0$ and $c_2 \approx -40$; show that both corrected gradients equal $\nabla f = -20$.)

4. Does FedAvg with full participation and exact gradients necessarily converge to the global optimum for convex functions with $E > 1$ and a constant learning rate? Using the example of item 2, find the fixed point of the iteration and state the conditions that make convergence to the optimum possible.

(b) [3 points] Batch normalization and personalization.

1. Why is statistical averaging of BatchNorm across hospitals invalid? Explain why the activation distribution depends on the imaging protocol.

2. What do FedBN and personalization layers offer? What is shared across clients and what stays local?

3. Name one privacy cost of keeping BN statistics local, and explain why these statistics can carry sensitive information.

(c) [3 points] DP-SGD and privacy accounting.

1. Write DP-SGD: per-sample gradient, clipping with threshold C, summation, Gaussian noise, division by batch size. Under add/remove adjacency, compute the ℓ_2 sensitivity of the clipped sum; then compare it in one sentence for replacement adjacency.

2. Why does clipping with threshold C guarantee, under add/remove adjacency, that changing one sample changes the sum by at most C? Use the triangle inequality.

3. With $q = 0.01$, $\sigma = 1.1$, $T = 1000$ and $\delta = 10^{-5}$, compute an approximate Rényi differential privacy (RDP) bound at order $\alpha = 10$:

$$\epsilon_{\text{RDP}}(\alpha) \approx \frac{Tq^2\alpha}{2\sigma^2}, \qquad \epsilon = \epsilon_{\text{RDP}}(\alpha) + \frac{\log(1/\delta)}{\alpha - 1}.$$

Why should one minimize over several α?

4. If a rare subgroup has $k = 10$ patients, what does group privacy say about its budget? (For a simple pure bound assume $\delta = 0$; in the (ϵ, δ) case also explain the growth of δ.) What does this imply for fairness? (Hint: write the definition of add/remove versus replacement adjacency and explain why "$\epsilon = 1.7$" without an adjacency unit is not a claim.)

(d) [4 points] Privacy audit and fairness.

1. Design a membership inference attack: which shadow models, which loss/confidence feature, and which metric (e.g. TPR at FPR $= 1\%$) do you report?

2. State the difference between federated learning and differential privacy in two precise sentences. Is a federated model automatically protected against membership inference? Why?

3. Suppose the gradients of rare-disease patients are larger than the threshold C. Show that clipping shrinks their contribution to the aggregate gradient. How can this lead to worse performance on the rare subgroup?

4. Propose an experiment to trace the fairness–privacy frontier: vary $\epsilon \in \{0.5, 1, 2, 4, 8\}$ and report Dice per subgroup, attack AUC and communication cost. What result would convince you that using DP is responsible in this scenario?

(e) [2 points] Deployment decision. Between plain FedAvg and FedBN with personalization, and between accepting the privacy budget computed in (c) or stopping the project, what do you choose? In at most 6 lines, justify it with three numbers (the drift of (a), the ϵ of (c), and the group law for $k = 10$). An answer without all three numbers receives no credit.
$\star$ **[2 points].** With $q = 0.01$, $\sigma = 1.1$, $T = 1000$, $\delta = 10^{-5}$, the approximation $\epsilon_{\text{RDP}}(\alpha) = Tq^2\alpha/2\sigma^2$, and the conversion $\epsilon = \epsilon_{\text{RDP}}(\alpha) + \log(1/\delta)/(\alpha - 1)$, compute ϵ for $\alpha \in \{2, 10, 18, 32\}$. Where is the minimum? Show by differentiation that the optimal order follows from $Tq^2/2\sigma^2 = \log(1/\delta)/(\alpha - 1)^2$, and compute it.

Solution 5

(a)

1. The server sends w_t to K clients; client k performs E local steps with its local data of size n_k to obtain w_{t+1}^k; the server takes the weighted average:

$$w_{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_{t+1}^k, \qquad w_{t+1}^k = w_t - \eta \sum_{\tau=1}^{E} \nabla F_k(w_{t,\tau}^k).$$

2. For $F_k(w) = \frac{1}{2} a_k (w - b_k)^2$, after E steps:

$$w_k^{(E)} = (1 - \eta a_k)^E w_t + \left[1 - (1 - \eta a_k)^E\right] b_k.$$

With $a_1 = 1, b_1 = 0$, $a_2 = 4, b_2 = 10$, $\eta = 0.1$, $E = 2$ and equal weights:

$$w_1 = 0, \qquad w_2 = (0.6)^2 \cdot 0 + (1 - 0.36) \cdot 10 = 6.4, \qquad w_{\text{FedAvg}} = 3.2.$$

One global gradient step from $w = 0$:

$$\nabla F(0) = \tfrac{1}{2} \cdot 1(0 - 0) + \tfrac{1}{2} \cdot 4(0 - 10) = -20, \qquad w_{\text{global}} = 0 - 0.1(-20) = 2.0.$$

FedAvg therefore reaches 3.2 while one global step reaches 2.0; the drift is 1.2. Repeating the same round gives $w_{t+1} = 0.585 w_t + 3.2$ with fixed point $w_\infty = 3.2/(1 - 0.585) \approx 7.71$, while the global optimum is $w^\star = 8$. The gap comes from the local steps $E > 1$ and from heterogeneity.

3. SCAFFOLD keeps a control variate c_k per client and c at the server, and changes the local update to

$$w \leftarrow w - \eta \left(\nabla F_k(w) - c_k + c\right).$$

c_k approximates the difference between the local and global gradients; when all clients are identical, $c_k = c$ and the correction vanishes. In practice c_k is updated from the difference between local and global gradients, not from a closed formula. Numerical check with item 2: $c_1 \approx \nabla F_1(0) = 0$, $c_2 \approx \nabla F_2(0) = -40$, $c \approx \nabla f(0) = -20$; corrected client 1: $0 - 0 + (-20) = -20$, client 2: $(-40) - (-40) + (-20) = -20 = \nabla f$. Both clients follow the central gradient.

4. This example shows that with $E > 1$ and a constant rate the weight iteration does not necessarily reach the global optimum: the fixed point is 7.71, not $w^\star = 8$. Convergence to a neighbourhood of the optimum is guaranteed only under the standard assumptions of smoothness/convexity, a sufficiently small step and a heterogeneity bound; reaching exactly 8 requires $E = 1$, a decreasing rate, or variance reduction such as SCAFFOLD. Partial participation and strong heterogeneity need additional assumptions.

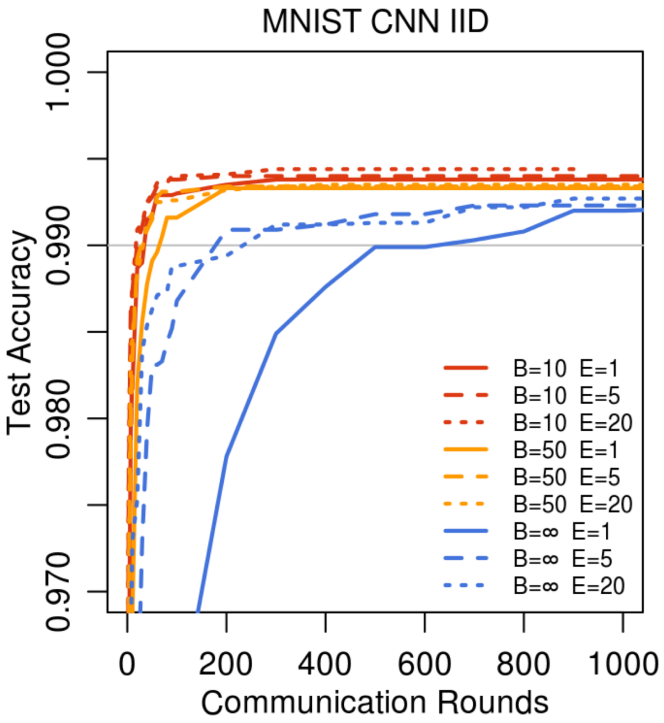


Figure: test accuracy versus communication rounds for IID and heterogeneous data. In the heterogeneous case convergence is slower and ends at a worse point — the same drift seen in the quadratic example (Fig. 2 of FedAvg).

(b)

1. BatchNorm estimates the mean and variance of activations on the local batch. The activation distribution depends on the imaging protocol, contrast, intensity and anatomy of each hospital; averaging these statistics across centers mixes heterogeneous distributions and can degrade performance.

2. FedBN keeps the BN layers local and shares the remaining weights. Personalization layers can keep the early layers shared and the later layers/head local. This absorbs heterogeneity, at the cost of communication and a possible loss of generalization.

3. BN statistics (mean/variance) can leak information about the local data distribution, the presence of subgroups, or even sample membership. Keeping them local reduces but does not remove the risk; if they are shared, they must be accompanied by DP or access restrictions.

(c)

1. For each sample i: $g_i = \nabla_\theta \ell(\theta, x_i)$; clipping:

$$\bar{g}_i = g_i \cdot \min\left(1, \frac{C}{\|g_i\|_2}\right);$$

then $\tilde{g} = \frac{1}{|B|}\left(\sum_{i \in B} \bar{g}_i + \mathcal{N}(0, \sigma^2 C^2 I)\right)$. Under add/remove adjacency the clipped sum changes by at most C, so the ℓ_2 sensitivity is C; for replacement the general bound is $2C$.

2. For every i: $\|\bar{g}_i\| \le C$. If g_i is replaced by g_i':

$$\|\bar{g}_i - \bar{g}_i'\| \le \|\bar{g}_i\| + \|\bar{g}_i'\| \le 2C.$$

The worst-case change of the sum is therefore C for add/remove and $2C$ for replacement. Accounting must keep this adjacency convention consistent with the noise and the RDP conversion.

3.

$$\epsilon_{\text{RDP}}(10) \approx \frac{1000 \cdot (0.01)^2 \cdot 10}{2(1.1)^2} = \frac{1}{2.42} \approx 0.413.$$

$$\epsilon \approx 0.413 + \frac{\log(10^5)}{9} \approx 0.413 + 1.279 = 1.69.$$

For $\alpha = 10$ this approximation gives $\epsilon \approx 1.69$; the final budget must be the minimum over all tested orders, i.e. about 1.42 at $\alpha = 18$ in the starred item.

4. Group privacy: for a group of k patients and pure privacy, $\epsilon_{\text{group}} \le k\epsilon$ (simple bound). With the final budget $\epsilon \approx 1.42$, for $k = 10$ this is about 14.2; if only $\alpha = 10$ is reported, the bound is 16.9. In approximate (ϵ, δ) privacy, δ also grows under group composition and must not be dropped. Adjacency note: add/remove (membership) differs from replacement (attribute/label).

(d)

1. For MIA: train several shadow models on subsets with and without the sample; for each sample take the loss or confidence feature; choose a threshold so that FPR is small (e.g. 1%); report TPR and AUC. A strong attack must show high TPR at low FPR, not only a high overall AUC.

2. FL is a training arrangement that does not move raw data; DP is a formal guarantee about outputs/updates. A federated model is not automatically protected against MIA; gradients and weights can leak information. DP must be applied and accounted for separately.

3. If $\|g_i\|_2 > C$, the clipping factor $C/\|g_i\|_2 < 1$ is smaller than one, and the sample's contribution to the sum shrinks by that factor. If rare-disease patients have larger errors and larger gradients, their contribution is systematically reduced; the model can therefore become worse on the rare subgroup. This is a fairness effect of uniform clipping.

4. Experiment: five hospitals, two rare subgroups, $\epsilon \in \{0.5, 1, 2, 4, 8\}$; for each ϵ: Dice per subgroup, attack AUC, communication cost and training time. A convincing result is a defensible operating point where the rare-subgroup drop is bounded and monitored, together with an independent audit and the possibility of stopping the model.

(e) The answer is graded. Rubric: half a point for an explicit architecture/training choice, half for the drift number of (a) (3.2 versus 2.0 or the optimum 8.0), half for the final budget $\epsilon \approx 1.42$ (or 1.69 at $\alpha = 10$ plus minimization), and half for the group law $k\epsilon \approx 14.2$ and the stop/continue decision for the rare subgroup. Without all three numbers, at most half credit. Sample full answer: "FedBN with personalization (not plain FedAvg, because the drift 1.2

and real heterogeneity break convergence); deploy only with $\epsilon \approx 1.42$ (minimum over orders) and monitor the $k = 10$ subgroup, for which the group law gives about 14.2; if the subgroup drops by more than 5 points, stop."

⋆ **answer.** $\epsilon_{\text{RDP}}(\alpha) = 0.0413\alpha$ and $\log(10^5) = 11.5129$: $\alpha = 2 \to 11.60$, $10 \to 1.69$, $18 \to 1.42$, $32 \to 1.69$. The minimum is near $\alpha = 18$; from $0.0413 = 11.513/(\alpha - 1)^2$, $(\alpha - 1)^2 \approx 278.6$ and $\alpha \approx 17.7$. Reporting a single α wastes about 0.27 units of budget. Rubric: one point for the table, one point for the optimality argument.

Problem 6. From error to foundation: error analysis and SAM-based segmentation [16]

Background. A hospital network wants to deploy a SAM-based tumor segmentation tool at three centers and compare it with a well-tuned nnU-Net. The mean Dice of the two models is almost identical, but physicians report that the type of errors differs: at one center the SAM tool misses the boundaries of small tumors; at another it extends the tumor into healthy tissue. This question asks you to design the error analysis and to decide, based on evidence, what to deploy.

Main sources: SAM, SAM 2, SAM 3, MedSAM, Medical SAM Adapter, MedSAM3, SAM3-Adapter, Comparing SAM 2 and SAM 3, False Promises, Aggregation Hides Failures. All concepts needed are stated in the text; the papers are pointers for deeper reading.

(a) [4 points] Stratified error analysis and Simpson's paradox. Suppose Dice is given by center and tumor size as follows:

Center	Large tumor (n)	Dice large	Small tumor (n)	Dice small
A	90	0.90	10	0.80
B	10	0.95	90	0.85

1. Compute the sample-weighted Dice mean and the macro mean (simple mean over subgroups) for each center. Explain that the weighted Dice mean is not a true micro Dice unless TP/FP/FN counts are also given.

2. Explain why center B is better in both subgroups but worse in the overall mean. What kind of paradox is this, and what does it imply for reporting a single number?

3. Propose a slice-level analysis design: which factors do you stratify by, how many samples per cell do you report, and which confidence interval do you use?

4. How do you measure inter-reader variability? Propose two ways to report multi-reader ground truth.

5. Consider two sites with $n = 1000$ and $n = 300$. At the first site model A is correct on $880/1000$ and model B on $860/1000$; at the second site A is correct on $150/300$ and B on $165/300$. Compute the weighted and macro means for both models and show that the winner flips. Why is reporting only a weighted mean dangerous for deployment?

(b) [3 points] SAM architecture and prompts.

1. Name the three main components of SAM and explain each: the image encoder (heavy, once per image), the prompt encoder (light, turns points/boxes/masks into tokens; the original paper also reports preliminary free-form text results, but the standard interface is mainly spatial), and the mask decoder (light, combines features and prompt).

2. For a 1024×1024 image with a ViT and patch size 16, compute the spatial dimensions of the image-encoder feature map. Why is the image encoder the computationally heavy part?

3. In SAM's multimask output mode, three masks and three IoU scores are produced. Why? What different meanings can a point on the lung have? How do you use the three masks at inference?

4. Consider one box and two points as a prompt. Explain how the prompt encoder turns them into tokens and why this part is light compared with the image encoder.

5. Give three reasons why SAM fails on medical images zero-shot (think of natural versus medical domain, resolution and contrast, and the difference between spatial prompts and free-form text, and the limits of the standard interface).

(c) [3 points] SAM 2, SAM 3 and extension to volumes.

1. How does SAM 2 extend promptable segmentation to video? Explain its three mechanisms: the memory encoder (compresses features of previous frames), the memory bank (stores them), and memory attention (combines them with the current frame).

2. If we treat the slices of a CT scan as video frames, which assumption about temporal continuity do we make? Why can a 5 mm slice spacing and anisotropy break this assumption?

3. What capabilities does SAM 3 add to SAM 1/2? Note that visual prompts (point/box/-mask) still exist. Explain the concept prompt and the presence head, and how they separate detection from localization.

4. Consider the phrase "enhancing tumor core" as a text prompt. Name two sources of ambiguity in this phrase and explain why a general text encoder may fail to ground it correctly.

(d) [4 points] Adaptation and fair comparison.

1. Consider four adaptations with this split: MedSAM (fine-tune the image encoder and mask decoder; prompt encoder fixed), Medical SAM Adapter (image encoder fixed, thin trainable adapters), SAM-Adapter (task-specific adapters on a fixed model), SAM3-Adapter (the same idea for SAM 3). For each, say what happens with little data and a small shift, and with abundant data and a structural shift; which do you recommend for our three-center deployment?

2. Compare full fine-tuning, linear probing and adapters in terms of the bias–variance tradeoff and catastrophic forgetting: when is a small adapter sufficient, and when is full fine-tuning necessary?

3. Design a fair comparison between adapted SAM and nnU-Net. Mention at least: data split, prompt budget, tuning budget, inference cost, and out-of-center evaluation.

4. Propose a deployment decision rule: when nnU-Net, when adapted SAM, and when a human-in-the-loop combination is appropriate?

(e) [2 points] Deployment gate. Deploy only if all three conditions hold: (i) a stratified report (weighted and macro) with confidence intervals, not a single mean; (ii) equal prompt and tuning budgets for the two models; (iii) a human-in-the-loop path for uncertain

cases. In at most 6 lines, justify it with one number from (a) (weighted/macro) and one condition from (d). An answer without a number receives no credit.

$\star$ **[1 point].** Write the condition for weighted/macro inversion in one equation ($\arg\max$ weighted $\neq$ $\arg\max$ macro) and explain, using the site weights (1000 versus 300), why it occurred in part (a).

Solution 6

(a)

1. Sample-weighted Dice for A:

$$\frac{90 \cdot 0.90 + 10 \cdot 0.80}{100} = 0.89,$$

and macro: $(0.90 + 0.80)/2 = 0.85$. For B:

$$\frac{10 \cdot 0.95 + 90 \cdot 0.85}{100} = 0.86,$$

and macro: $(0.95 + 0.85)/2 = 0.90$.

2. B is better in both subgroups ($0.95 > 0.90$ and $0.85 > 0.80$) but worse in the overall mean ($0.86 < 0.89$), because the share of small tumors is larger in B. This is an instance of Simpson's paradox. The message: a single number can reverse when the subgroup mix changes; report the n of each cell and both means.

3. Slice-level design: stratify by center, tumor size (small/medium/large), image quality, scanner and reader; report n, Dice, HD95, lesion sensitivity and a bootstrap confidence interval per cell; use paired tests between models within each cell.

4. Inter-reader variability: pairwise Dice between readers, agreement coefficient, and variance. Two options: (i) multiple readers and report model performance against each reader; (ii) soft/probabilistic labels and distributional evaluation.

5. Weighted Dice: A with $(880 + 150)/1300 = 1030/1300 \approx 0.792$ and B with $(860 + 165)/1300 = 1025/1300 \approx 0.788$; A wins by $+0.004$. Macro: A with $(0.88+0.50)/2 = 0.69$ and B with $(0.86 + 0.55)/2 = 0.705$; B wins by $+0.015$. The winner flips because the site weights (1000 versus 300) act against the direction of the simple mean. Reporting only a weighted mean is dangerous because the large site imposes the majority view and buries a failure at the small site (which may be exactly our deployment center).

(b)

1. **Image encoder:** a heavy ViT that turns the image into a dense feature map. **Prompt encoder:** turns points, boxes and masks into light tokens; the SAM paper also reports preliminary free-form text-prompt results, but the standard and common interface is spatial. **Mask decoder:** a light transformer with bidirectional cross-attention between prompt tokens and image features, producing the masks.

2. With patch 16 and image 1024, the number of tokens is 64×64. The channel count (256) follows only under a specified neck assumption; under that assumption the map is $64 \times 64 \times 256$. The image encoder is heavy because it has millions of parameters and global attention over thousands of tokens.

3. In the multimask output mode, three masks resolve ambiguity: a point on the lung can mean the whole lung, one lobe, or a lesion. The model produces three interpretations at different scales and gives each an IoU score. At inference one can take the highest score or let the user choose.

4. In the standard implementation, a box is represented by two corner points (top-left and bottom-right) and two points by two tokens; hence four point tokens (plus a mask token if present). The prompt encoder is light because it processes only a few tokens, whereas the image encoder has thousands.

5. Zero-shot failure reasons: (i) natural versus medical domain (texture, contrast, modality); (ii) resolution/scale and small lesions; (iii) free-form text prompts (in experimental paths or text-capable models such as SAM3) versus clinical terminology; (iv) the difference between the SA-1B distribution and hospital data; (v) the three-dimensional/multi-frame nature.

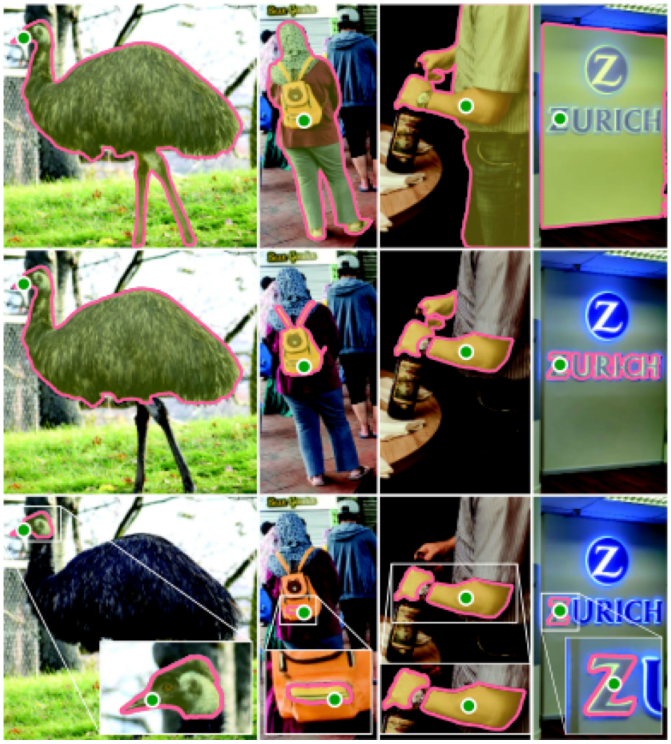


Figure: three valid masks from one ambiguous point. A single-output average gives a grey boundary, which is why SAM is trained as a multi-hypothesis model with an IoU-score head (Fig. 3 of SAM).

(c)

1. SAM 2 has a streaming memory: the memory encoder compresses features of previous frames; the memory bank stores them; memory attention combines the current frame's features with the memory so that the mask stays stable over time.

2. Treating slices as frames assumes that changes between slices are small and continuous. A 5 mm spacing and anisotropy can cause anatomical jumps; the memory may propagate the mask to the wrong structure in the next slice. Remedies: slice interpolation, closer spacing, or a 3D adapter.

3. SAM 3, in addition to the visual prompts of previous generations, accepts a concept prompt (a short noun phrase or an image exemplar) and finds all instances of that concept. The presence head decides whether the concept is present at all, separating this decision from localization; this reduces false positives caused by absent objects.

4. Two sources of ambiguity: (i) "enhancing tumor core" can refer to the enhancing region in MRI or to active tumor tissue in CT; (ii) the boundary between tumor and edema/necrosis is ambiguous. A general text encoder may fail to map the term to the correct visual concept; a clinical vocabulary or domain training is needed.

(d)

1. MedSAM (fine-tune image encoder and mask decoder, prompt encoder fixed): with abundant data and a structural shift it has the best ceiling, but with little data it overfits and forgets. Medical SAM Adapter and SAM-Adapter (adapters on a fixed model): optimal with little data and a small/local shift; general knowledge is preserved and training cost is low. SAM3-Adapter: the same for concept prompts. For a three-center deployment with limited labels: adapter + external evaluation; full fine-tuning only if enough data and a structural shift (new modality) exist.

2. Full fine-tuning has the lowest bias but the highest variance and the risk of catastrophic forgetting of general knowledge; linear probing (head only) has the lowest variance but high bias (natural features fixed for medicine); the adapter is the middle point: low capacity to absorb domain shift without forgetting. An adapter suffices when the domain gap is small/local (contrast, texture) and target data are scarce; full fine-tuning is needed when the gap is structural (new modality, different anatomy) and enough data exist to control variance.

3. Fair comparison: identical patient-level and center-level split; identical prompt budget (e.g. one box or three points); identical tuning budget (GPU hours, hyperparameter search); inference cost and memory; evaluation on an external center; report Dice, HD95, ASSD, time and memory; compare with nnU-Net trained on the same data. Comparing zero-shot SAM with a fully trained nnU-Net is not fair.

4. Decision rule: if dense labels and high accuracy are required and centers are homogeneous, nnU-Net; if labels are scarce and a user can provide prompts, adapted SAM; if rare/open structures and text descriptions are needed, SAM 3/concept with human-in-the-loop; in all cases, error monitoring and external evaluation are conditions for deployment.

(e) The answer is graded. Rubric: half a point for the stratified report (weighted and macro) with one number from (a) ($0.89/0.86$ or the inversion $0.792/0.788$ versus $0.69/0.705$), half for equal prompt and tuning budgets, half for the human-in-the-loop condition, and half for an explicit deploy/no-deploy decision. Without a number, at most one point. Sample full answer: "do not deploy until the stratified report is complete; the weighted mean 0.792 versus 0.788 makes A the winner and the macro 0.69 versus 0.705 makes B the winner, so one number cannot decide; deployment is conditional on equal prompt/tuning budgets and human-in-the-loop for uncertain cases."

$\star$ **answer.** Condition: arg max weighted $\neq$ arg max macro. Here the weight of the large site ($1000/1300$) imposes the majority view on the weighted mean (A by $+0.004$), while the simple mean gives each site an equal vote (B by $+0.015$). Inversion can occur whenever the sampling weights are not aligned with the direction of superiority. Rubric: half a point for the equation, half for the weight argument.